

LAB 3: Propositional Logic & Model Checking Group Challenge

Instructor	Dr. Naila Marir	Duration	1 hour 30 minutes
Semester	Fall 2026	Group size	3–4 students
Submission	One GitHub link per group	Total marks	100

Group No.	2		
Member name	Student ID	Member name	Student ID
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1. Represent facts as propositional logic sentences using Not, And, Or and Implication. 2. Explain how `model_check` decides whether a knowledge base entails a query. 3. Encode a new logic puzzle in Python and solve it with model checking.

1. Python 3 and the package `termcolor` (install with `pip install termcolor`).
2. Lab files (on Blackboard): `logic.py`, `harry.py`, `clue.py`, `puzzle.py`, `mastermind.py`. Put all of them in one folder.
3. A GitHub account for each group.

Step	Activity	Time
0	Setup: run <code>harry.py</code> and <code>clue.py</code>	5 min
1	Task 1 — Understand the code	20 min
2	Task 2 — Change the knowledge	20 min
3	Task 3 — Knights & Knaves challenge	30 min
4	Write the PDF and push to GitHub	15 min

1. **harry.py**: Write the truth table for rain, hagrid, dumbledore (8 rows). Mark the rows where the knowledge base is true. Use the table to explain why the program prints True.

R	H	D	$\sim R \rightarrow H$	$H \vee D$	$\rightarrow (H \square D)$	D	KB (all 4)
F	F	F	T	F	T	F	F
F	F	T	F	T	T	T	F
F	T	F	T	T	T	F	F
F	T	T	T	T	F	T	F
T	F	F	T	F	T	F	F
T	F	T	T	T	T	T	T
T	T	F	T	T	T	F	F
T	T	T	T	T	F	T	F

2. **clue.py**: The program prints MsScarlet, library and knife. Explain in plain English why each one must be true.

The murderer must be one of {mustard, plum, scarlet}. We are told Not(mustard) and Not(plum). The only person left is scarlet. Therefore MsScarlet is the murderer.

The crime happened in one of {ballroom, kitchen, library}. We are told Not(kitchen) and Not(ballroom). The only room left is the library. Therefore library is the room.

The weapon is one of {knife, revolver, wrench}. We are told Not(revolver). The hint says: at least one of {Not(scarlet), Not(library), Not(wrench)} is true. But we already proved scarlet is the murderer and library is the room. So Not(scarlet) and Not(library) are false. For the hint's Or to be true, the remaining Not(wrench) must be true. So the weapon is not wrench. Combined with Not(revolver), the only weapon left is knife. Therefore knife is the weapon.

For each change: **(a)** predict the result, **(b)** run the code, **(c)** explain the result.

1. **harry.py**: Delete the last fact dumbledore from the knowledge base. What does the program print now? Why?

We predict that the program will print false because there is ambiguity. The program result printed false. After removing dumbledore, the knowledge base no longer considers rain to be true because it has other cases where it can be false while the KB is true.

```
PS C:\Users\yui\z\Desktop\AI> ^C
PS C:\Users\yui\z\Desktop\AI>
PS C:\Users\yui\z\Desktop\AI> cd 'c:\Users\yui\z\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\yui\z\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '55802' '--' 'c:\Users\yui\z\Desktop\AI\labs\lab2\harry.py'
False
PS C:\Users\yui\z\Desktop\AI> 
```

2. **puzzle.py**: Delete the clue `MinervaGryffindor`. Who can still be placed in a house? Then add **one new clue of your own** that makes the answer unique again.

Deleting `MinervaGryffindor` removes the only fact tying `Minerva` to a house, so many assignments become possible, and the program prints nothing. The program results in no output. The KB no longer forces any positive placement, which makes no single placement certain, so it doesn't give anything for the program to print.

```
PS C:\Users\yuizs\Desktop\AI> ^C
PS C:\Users\yuizs\Desktop\AI>
PS C:\Users\yuizs\Desktop\AI> c:: cd 'c:\Users\yuizs\Desktop\AI'; & 'C:\Users\yuizs\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\yuizs\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '56379' '--' 'c:\Users\yuizs\Desktop\AI\labs\lab2\puzzle.py'
PS C:\Users\yuizs\Desktop\AI> 
```

6. Task 3 — Group Challenge: Knights & Knaves (30 marks, 30 min)

On an island, a **knight** always tells the truth and a **knave** always lies. Every person is one or the other, never both. Create a new file `knights.py` that imports `logic.py` and solves the puzzles below.

Hint — use this pattern for what a person says:

```
Implication(AKnight, statement) # a knight's statement is true
Implication(AKnave, Not(statement)) # a knave's statement is false
```

1. **Puzzle 0**: A says “I am both a knight and a knave.”
2. **Puzzle 1**: A says “We are both knaves.” B says nothing.
3. **Puzzle 2**: A says “We are the same kind.” B says “We are of different kinds.”

Bonus (+10 marks) — Puzzle 3: A says either “I am a knight.” or “I am a knave.” (you don't know which). B says “A said ‘I am a knave’.” B also says “C is a knave.” C says “A is a knight.”

Code is in `knights.py` file

```
Problems  Output  Debug Console  Terminal  Ports
bs\lab2\knights.py'
Puzzle 0:
A is a Knave

Puzzle 1:
A is a Knave
B is a Knight

Puzzle 2:
A is a Knave
B is a Knight

Puzzle 3 (Bonus):
A: unknown
B is a Knave
C: unknown
```

Create **one public GitHub repository** named Lab3_GroupX and submit **only its link** on Blackboard.
The repository must contain:

1. Lab3_GroupX.pdf — group number, member names, and for each task: your answer, your code, and a screenshot of the output.
2. All .py files — logic.py, your changed harry.py and puzzle.py, and your new knights.py. 3. README.md — member names and one line on how to run knights.py.

Before you submit: open your link in a private (incognito) browser window to check that it works.

Task	What is checked	Marks
Task 1	Correct truth table and clear explanations	30
Task 2	Predictions, results and explanations; a working new clue	30
Task 3	Puzzles 0–2 encoded correctly with output and explanation	30
Submission	Clear PDF and complete GitHub repository	10
Total		100
Bonus	Puzzle 3 solved and explained	+10